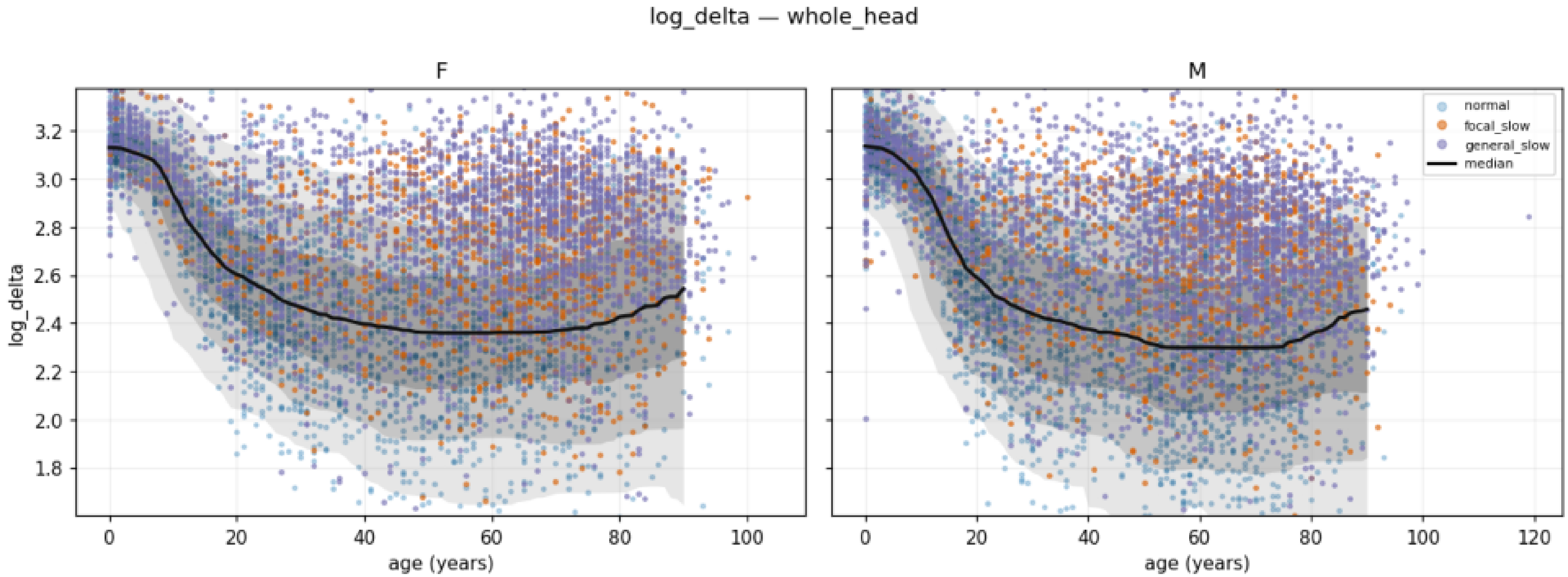
EEG SLOWING GROWTH CURVES

Age × sex normative percentile curves for quantitative EEG slowing

Built on 12,379 recordings (normal 4,916 · focal 2,067 · generalized 5,396). Curves show normal percentiles (median + 25-75 / 10-90 / 3-97 bands); focal and generalized recordings overlaid.
v1 — stage-agnostic.

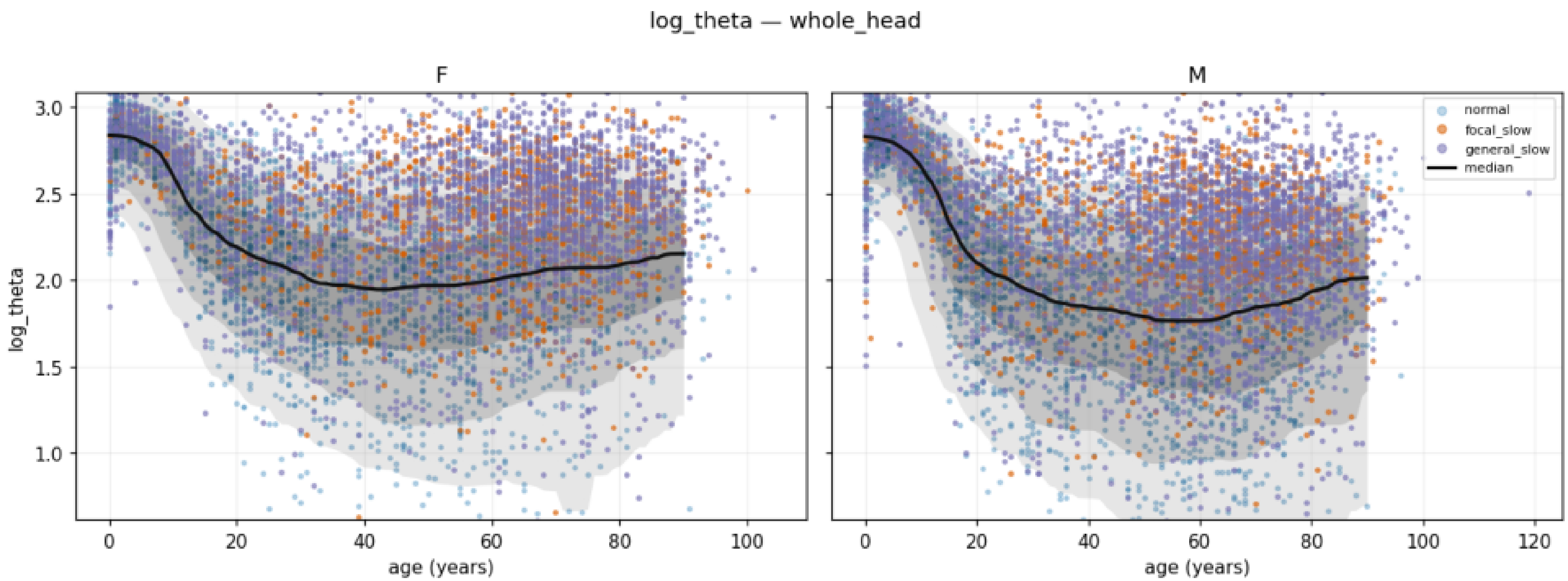
0.75	0.03	delta · theta · TAR
absolute delta AUC (adj)	normal median z	top features

Absolute delta power (log) — whole head



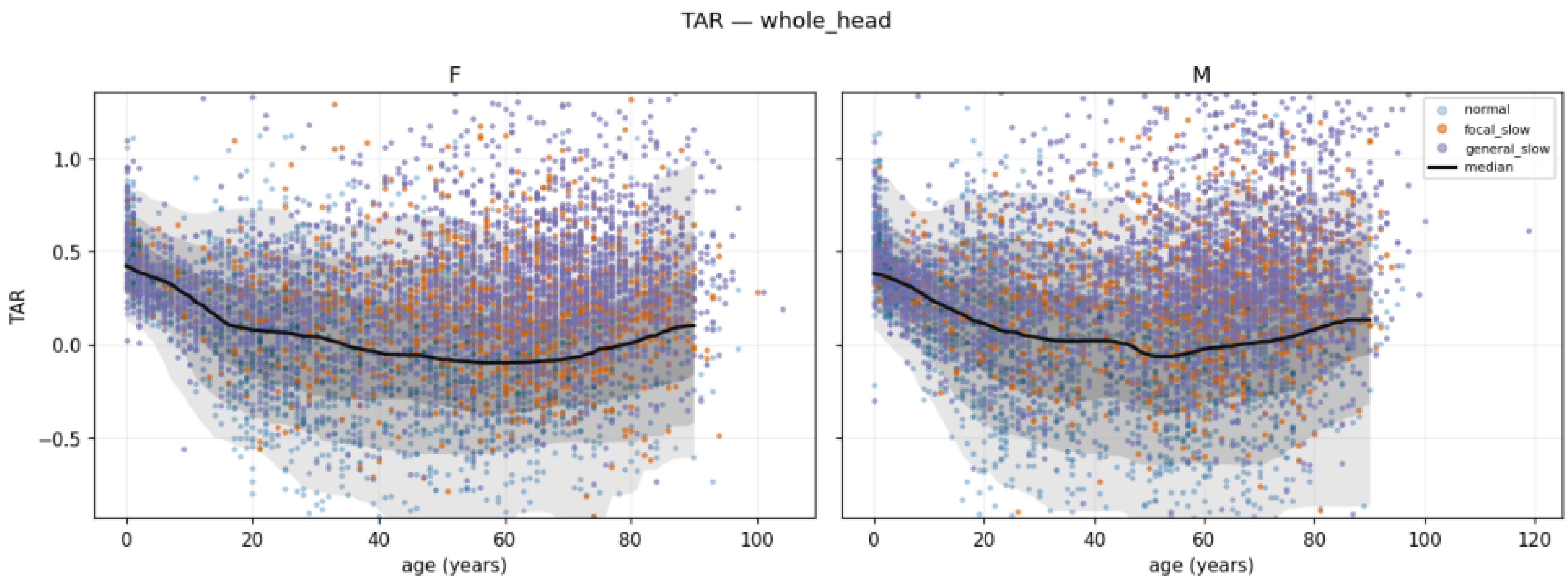
Validation figure. Normal median (black) falls steeply through childhood to a plateau by ~30 — the textbook maturational trajectory. Focal (orange) and generalized (purple) sit above the normal band. Strongest single discriminator: AUC 0.75 (age/sex-adjusted).

Absolute theta power (log) — whole head



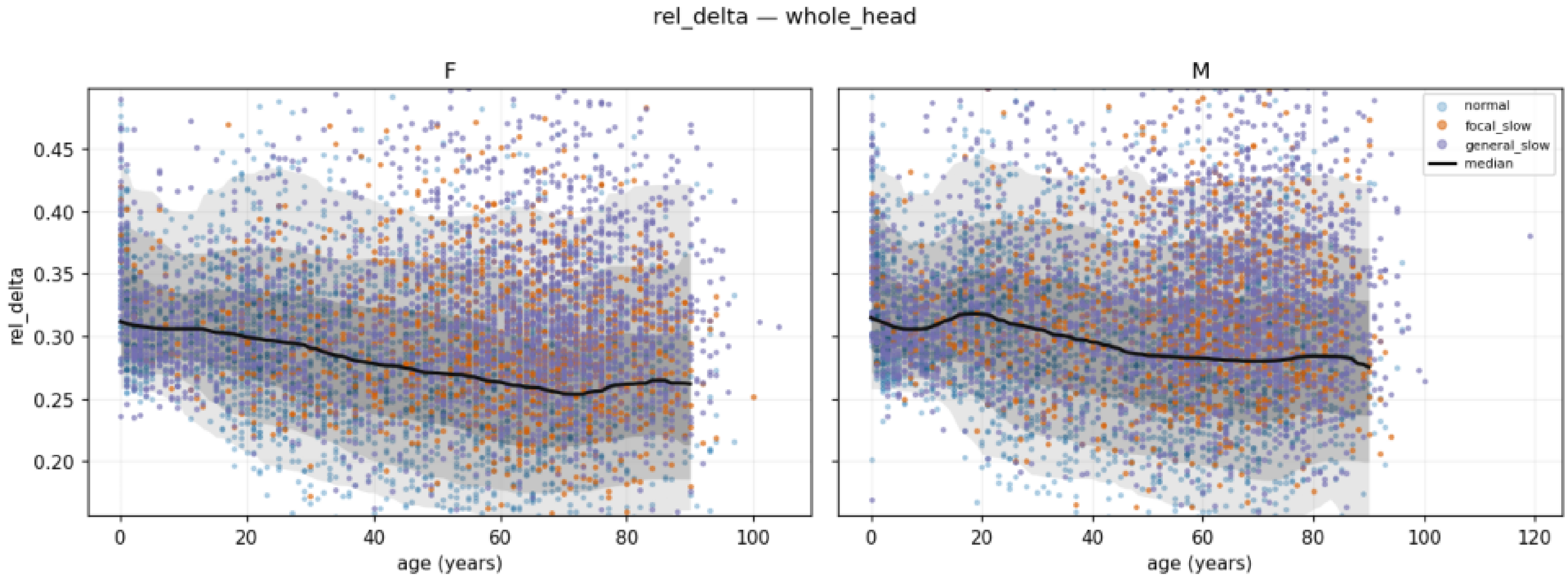
Second-strongest marker (AUC ~0.74). Slowing groups elevated above the age-conditioned normal band.

Theta/alpha ratio (TAR) — whole head



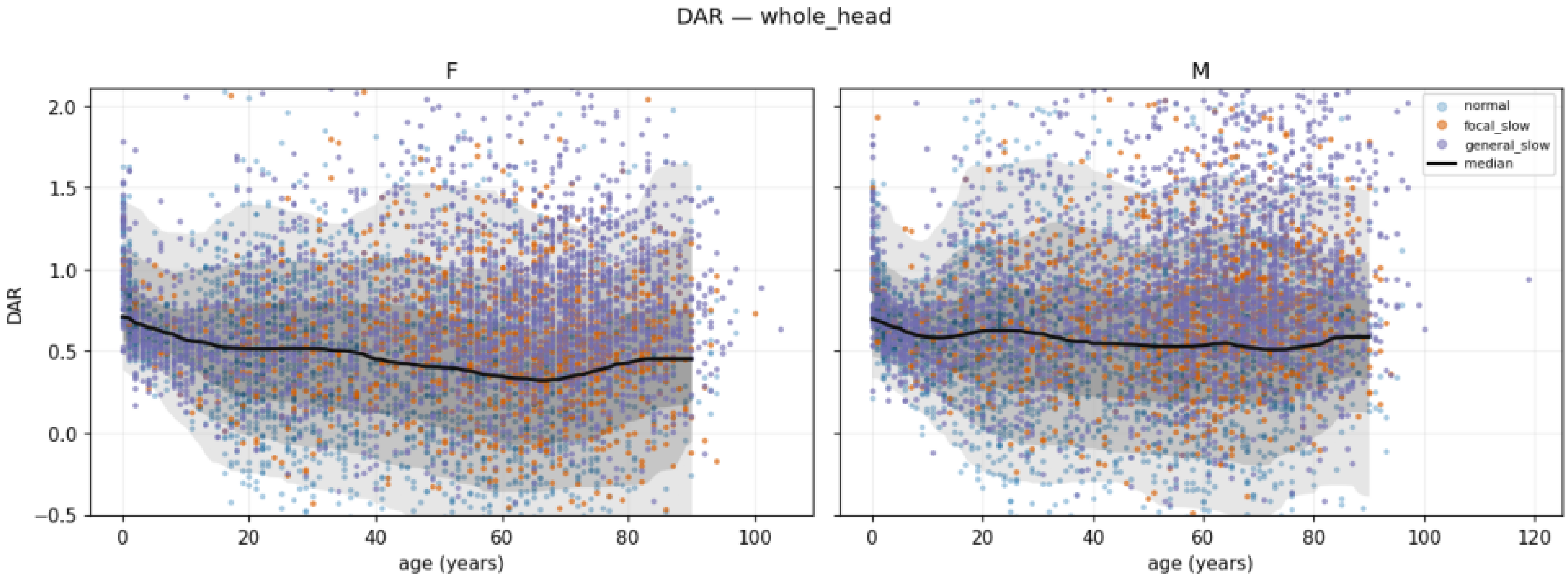
Spectral-shift ratio; strong even before adjustment (raw AUC ~0.71) — relative loss of faster (alpha) activity.

Relative delta power (delta/total) — whole head



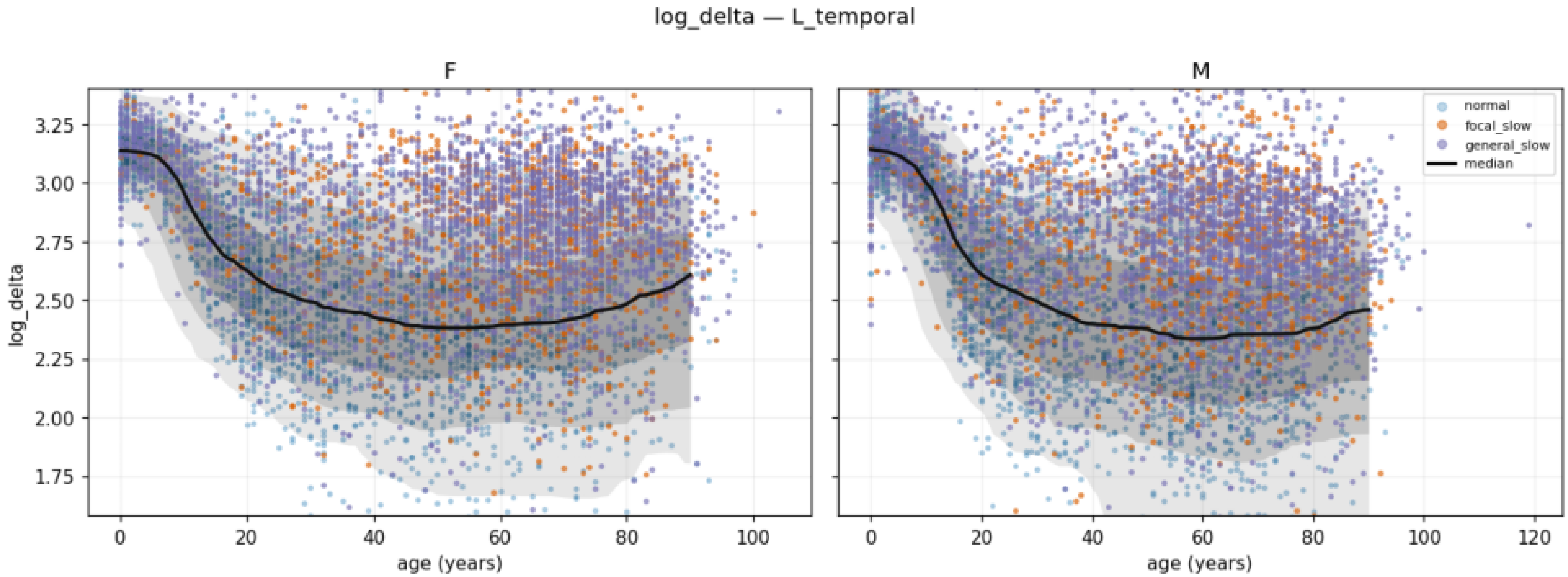
Delta as a fraction of total power. Weaker than absolute delta in v1 — pooling sleep stages dilutes the signal; stage stratification (v2) should sharpen it.

Delta/alpha ratio (DAR) — whole head



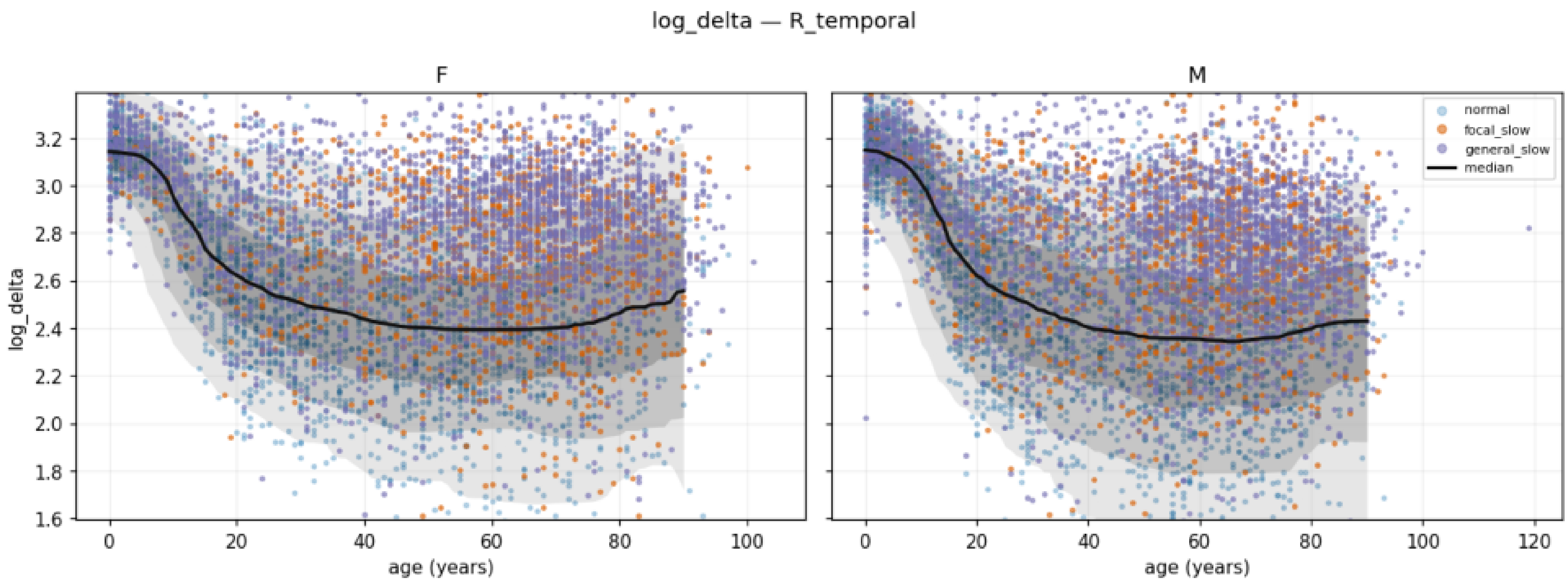
Classic slowing index; elevated in both slowing groups.

Absolute delta — left temporal



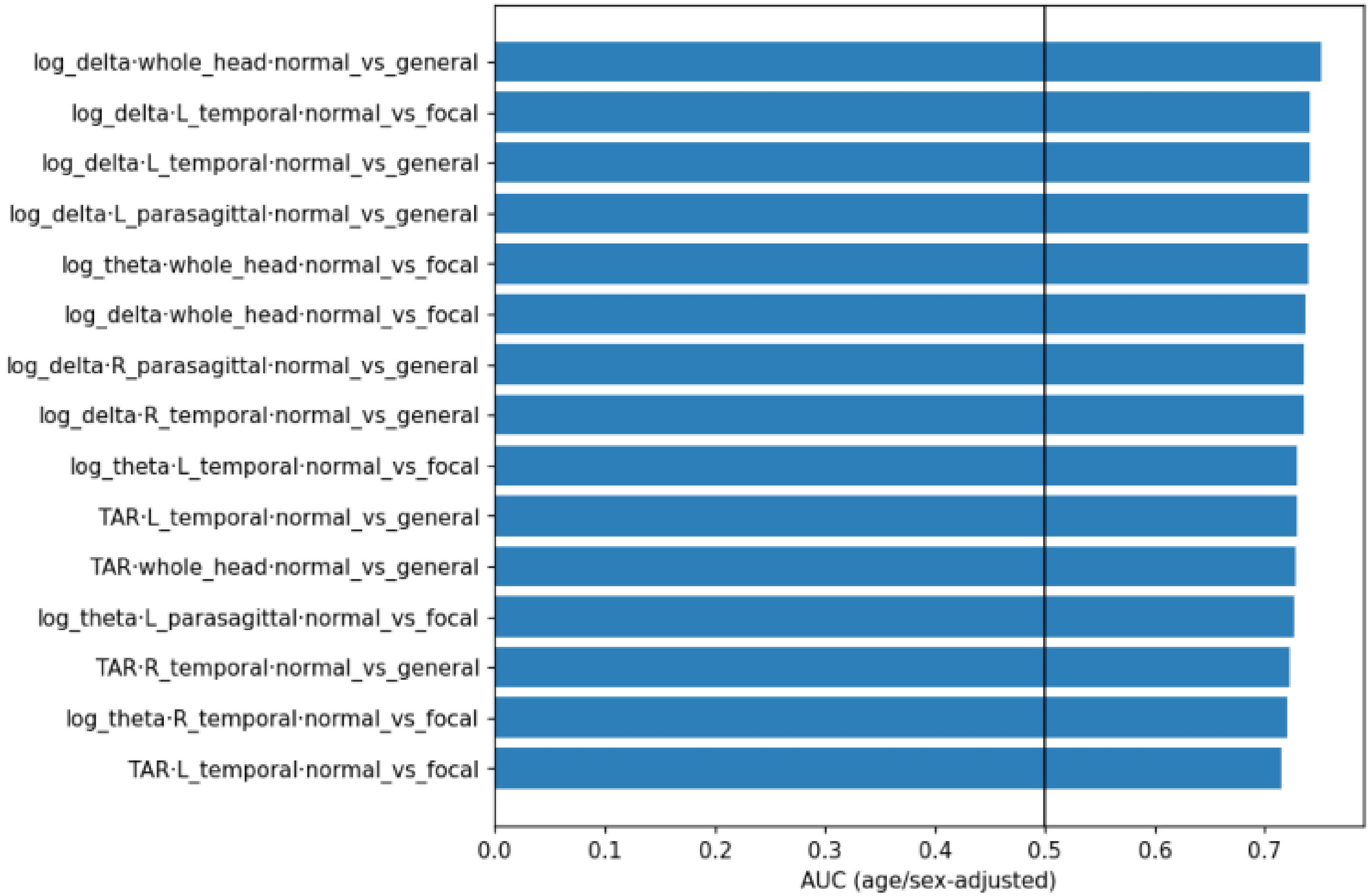
Regional view. Focal cases can deviate in one region while whole-head stays near normal — basis for topographic classification.

Absolute delta — right temporal



Right-temporal counterpart. Left-right differences feed the asymmetry score (focal vs generalized).

Top discriminative features (age/sex-adjusted AUC)



Every feature x region x group-pair ranked by AUC of its normal-referenced z. Absolute delta/theta power and TAR lead; 0.5 = no signal.